Milestone 3 - Classifier Build and Evaluation

Natalia Kusmirek | Readings in Computer Science | Instructor: Prof. Maira Marques Samary

1. Data, models, and evaluation protocol

Seed data. seed_dataset.csv: 524 citations from 156 distinct hosts, labeled with the six Milestone 2 categories. Labels came from an AI-assisted first pass (Profound's own citation category mapped onto the taxonomy, plus hand-built domain lists, then corrected) that the student reviewed and confirmed; they are not fully independent hand labels. Data are concentrated: www.bc.edu alone is 165 of the 174 BC-owned rows, and several categories have few hosts (hosts per category: BC-owned 5, earned media 18, education directory/ranking site 34, forum/community 9, competitor institution 65, other 25).

Public training signal. Curlie/DMOZ could not be used: curlie.org is unreachable from the build environment and no maintained public dump exists. Substitutes, mapped onto the taxonomy: Hipo US-university domains -> competitor institution (300 sampled); UT1 forums + social-network lists -> forum/community (300); UT1 press list (.com/.org/.net) -> earned media (300); UT1 job-search + financial lists -> other (300). Sampling is capped at 300 per category and public rows are weighted 0.3. No public source exists for BC-owned or education directory/ranking site, so those two categories learn from seed data only. Any public domain sharing a registered domain with a seed host is dropped so public data cannot leak a seed label.

Models. (1) *Rule baseline*: ordered keyword/domain-pattern rules (bc.edu suffix, social/forum names, .edu suffix, ranking/admissions keywords, news keywords, else other). (2) *Learned*: multinomial logistic regression (C = 10, fixed in advance, class-balanced) on TF-IDF character 2-5-grams of the host plus word tokens of the URL path. (3) *BC rule + LR*: because BC ownership is definitional (taxonomy rule: bc.edu and subdomains) and one host dominates the class, a hard ownership rule handles BC-owned and the LR is trained on the other five categories. Each learned variant is run with and without public data; the pure LR variants are kept in Table 2 so the effect of the rule is visible.

Protocol. Every split is grouped by host, so no domain appears in both train and test. Primary result: one stratified grouped hold-out of 93 rows (17.7% of the data, 37 hosts; the split was picked for size and category coverage, never by model performance; it is just under 20% because www.bc.edu must sit in train). Robustness: 5 x 5-fold grouped cross-validation (CV). The better-performing model is the one with the highest mean CV accuracy, a rule fixed before any results were seen. For contrast, a naive random row split of the same learned model scores 91.4%, versus 72.3% grouped: shared domains inflate accuracy by about 19 points.

2. Results

Category	**Test rows / hosts**	**Rule baseline P / R / F1**	**BC rule + LR, seed + public P / R / F1**
BC-owned	6 / 1	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
earned media	6 / 5	1.00 / 0.17 / 0.29	0.17 / 0.17 / 0.17
education directory/ranking site	31 / 7	0.88 / 0.90 / 0.89	0.85 / 0.94 / 0.89
forum/community	15 / 3	0.94 / 1.00 / 0.97	0.75 / 0.60 / 0.67
competitor institution	26 / 17	0.80 / 0.92 / 0.86	0.81 / 0.96 / 0.88
other	9 / 4	0.25 / 0.22 / 0.24	0.25 / 0.11 / 0.15
Overall accuracy	**93 / 37**	**81.7%**	**76.3%**
Macro-F1		**0.706**	**0.626**
Accuracy excluding BC-owned rows		**80.5%**	**74.7%**

Table 1. Per-category precision, recall, F1 on the primary grouped hold-out. Supports are tiny (BC-owned is 1 host, forum/community 3 hosts, earned media 5 hosts), so single-split per-category values are fragile; CV (Table 2 and Section 3) is the more reliable guide.

Model	Primary split accuracy	CV accuracy (mean +/- sd)	CV accuracy excl. BC-owned	CV macro-F1
Rule baseline (selected)	81.7%	88.4% +/- 0.0	82.6%	0.746
LR, seed only	69.9%	72.3% +/- 0.9	62.9%	0.510
LR, seed + public	75.3%	46.5% +/- 0.6	67.9%	0.429
BC rule + LR, seed only	69.9%	74.8% +/- 0.8	62.3%	0.515
BC rule + LR, seed + public	76.3%	78.7% +/- 0.5	68.2%	0.655

Table 2. Overall comparison (5 x 5-fold grouped CV). The baseline's sd is 0 because it has no trained parameters. Pure-LR accuracy with public data is low overall because, when bc.edu is held out, public .edu rows outvote the four tiny BC subdomains (BC-owned F1 0.07); the ownership rule removes that artifact.

3. Does it meet the 80% target? Analysis and limitations

Target. The rule baseline reaches 81.7% on the primary hold-out and 88.4% in CV, above 80%; it is therefore the selected model. The best learned model (BC rule + LR, seed + public) reaches 76.3% and 78.7%, short of 80%. The baseline's lead should not be read as rules beating learning: its keyword lists were written after looking at these same domains, and the seed labels were built with similar domain lists, so its numbers are optimistic and not a true held-out measurement. The fair head-to-head is the unseen-domain check below.

Why the learned model falls short. (1) Little data: 156 hosts, with only 18 earned-media and 25 other hosts, so character n-grams cannot learn a category from a handful of names. (2) Earned media vs other vs directory is a question about what a site does, which a domain string often does not reveal (e.g. catholic-link.org, empowerly.com); the dominant confusions are directory -> other and earned media -> other. (3) The public lists are domain-only and noisy (UT1's press list is dominated by small local newspapers, not the outlets AI engines cite) and cover four of six categories. (4) Seed labels are partly heuristic, so the model also learns labeling quirks. Even so, public data helps: BC rule + LR rises from 74.8% to 78.7% CV accuracy and macro-F1 from 0.515 to 0.655, mainly through forum/community (F1 0.26 to 0.76), since unseen social platforms look nothing like the seed ones by name. Directory/ranking F1 dipped slightly (0.73 to 0.69): it has no public source, and with public data its sites were more often labeled earned media or other (errors.csv).

Weak categories (both models). Earned media (CV F1 0.23 baseline / 0.25 learned) and other (0.47 / 0.29) are the problem classes. Candidate fixes for Milestone 4/5: add page-title and snippet text (already in the Profound export) as features, label more domains, and add a cheap lookup for well-known news outlets.

Manual check (30 unseen domains). The student reviewed 30 citations drawn from 284 domains that appear in the Profound export but not in the seed set (up to 5 per predicted category; the sample is not proportional to the population). Agreement with the selected model (Rule baseline): **22/30 = 73.3%**. Same 30 against each model: Rule baseline 73.3%; LR, seed only 66.7%; LR, seed + public 73.3%; BC rule + LR, seed only 66.7%; BC rule + LR, seed + public 73.3%. Disagreements: www.univstats.com (predicted other, you: education directory/ranking site); www.careervillage.org (predicted other, you: forum/community); sjcadmissionsblog.com (predicted education directory/ranking site, you: competitor institution); www.rncareers.org (predicted other, you: education directory/ranking site); www.onlineu.com (predicted other, you: education directory/ranking site); community.pepperdine.edu (predicted forum/community, you: competitor institution); dev.comewander.ca (predicted education directory/ranking site, you: other); journals.plos.org (predicted earned media, you: other).

Other limitations. One hold-out of 93 rows is a small test set; CV repeats reuse the same 156 hosts, so the reported +/- reflects split randomness, not sampling uncertainty about new BC citations. Confidence for the rule baseline is a placeholder (1.0 rule fired, 0.5 fallback); classified_output.csv therefore also carries the best learned model's probabilities. classified_output.csv holds out-of-fold predictions, so no citation is scored by a model that saw its own domain's label.